Lessons Learned

furlink

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Introduction

The purpose of the lessons learned document for the FurLink project is to progressively capture project insights in a formal document for use by future SoCIT (School of Computing and Information Technologies) PBL groups and student developers on similar projects. This document will detail the challenges that arise throughout the development cycle of the previous versions, how the team navigates them, and how they can be avoided in the future. Additionally, it highlights what goes well, such as early budget optimization strategies, so that other project managers may adopt these successful practices. Upon project completion, this document will be formally communicated to the Asia Pacific College for project advisers and will become a part of the academic and organizational archives.

Lesson Learned Approach

The lessons learned for the furLink project are continuously compiled from Agile sprint retrospectives, scheduled standups, GitHub/Vercel deployment logs, and OpenProject logs throughout the project lifecycle. Insights will also be actively drawn from both realized and unrealized risks monitored in the project risk register, specifically focusing on the integrations with the Gemini 2.5 Flash AI and the PayMongo payment gateway. Ultimately, the findings documented here will be categorized by project knowledge areas. These knowledge areas consist of risk management, cost management, scope management, stakeholder management, and quality management.

Lessons Learned from this Project

The following chart lists the lessons being learned for the furlink project. It is important to note that not only failures or shortcomings are included but successes as well.

Category	Issue Name	Problem/Success	Impact	Recommendation
Risk Management	External API Dependencies	Success: The team identified the risk of PayMongo webhook delays and Gemini AI rate limits early in the planning phase.	By allocating a dedicated, isolated testing sprint specifically for the payment gateway, the team avoided deployment blockers and maintained the	Always allocate a dedicated buffer sprint for testing external third-party APIs before attempting to integrate them into the system workflow.

			August completion schedule.	
Scope & Security Management	AI Prompt Injection & Data Privacy	Problem: Initial AI integration tests revealed the Gemini Assistant could potentially be manipulated into accessing private business revenue tables.	Required an immediate architectural shift to implement Read-Only Database Views and strict Row-Level Security (RLS) to physically isolate the AI from sensitive data.	Never provide an LLM direct access to raw database tables. Always establish Read-Only Views and apply metadata filtering before integrating AI tools.
Quality Management	Pre-Deployment Code Stability	Problem: Late-stage code commits prior to integration testing caused serverless function cold-start delays.	Resulted in temporary performance drops on the Service Provider dashboard, requiring rapid Edge Function optimizations just days before the v1.0 launch.	Enforce a strict 48-hour "Code Freeze" prior to any major System/Integration testing window to ensure the testing environment is absolutely stable.
Stakeholder Management	Digital Onboarding	Problem: Traditional grooming shops struggled with the initial self-service digital onboarding process.	There may user adoption during the MVP phase, as shop owners were unaccustomed to configuring digital profiles and liability waivers.	When digitizing traditional local businesses, do not rely purely on automated onboarding. Offer trainings to lower the barrier to entry.

Lessons Learned Knowledge Base / Database

The lessons learned for the Furlink Project will be contained in the organizational lessons learned knowledge base maintained by the APC-SoCIT PBL Repository. This information will be cataloged under the project's year (2026) and the technology stack for future reference. This information will be valuable for any future student project manager assigned to a web-based scheduling or AI-integrated capstone project.

Lessons Learned Applied from Previous Projects

The Furlink Project used several lessons learned from the past version of furlink:

- The decision to use Supabase (PostgreSQL) over Firebase was based on the note that handling complex relational data (like overlapping calendar appointments) was highly inefficient and expensive in NoSQL environments.
- The implementation of a strict 48-hour code freeze before Final Testing was adopted based on Agile ceremonies. Not doing so will cause the project to have severe delays due to untested hotfixes breaking the production build hours before their defense.

Process Improvement Recommendations

As indicated in the lessons learned chart above, integrating Large Language Models (LLMs) like Gemini into direct database queries introduces severe security risks (Prompt Injection and Data Leakage) that standard web applications do not face. Not only is this a lesson learned for furlink, but the academic organization must ensure that all future project managers are aware of the need to secure their architectures against AI manipulation.

Therefore, it is recommended that APC-SoCIT standardize an "AI Security Checklist" for all future Capstone projects proposing LLM integrations. Operationally, future teams should mitigate adoption barriers by conducting on-site provider training and guided onboarding sessions. It is also recommended to implement real-time monitoring and buffer budgets for all AI services. Prior to development, teams must brief their project advisers on how they plan to implement Database Views and Restricted API Keys to make sure user data remains secure from AI related data leaks.

Sponsor Acceptance

Approved by the Project Sponsor:

____ Date: April 7, 2026

Manuel R. Doria Jr.

Project Sponsor Representative